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Annual Review of Animal Biosciences

A Passion for Small Things and Staying Primed: My Career in Virology and Immunology

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Abstract

A love of science and animals, perseverance, and happenstance propelled my career in veterinary virology and immunology. I have focused on deadly enteric and respiratory viral infections in neonatal livestock and humans with an aim to understand their prevalence, pathogenesis, interspecies transmission, and immunity and develop vaccines. Research on animal coronaviruses (CoVs), including their broad interspecies transmission, provided a foundation to understand emerging zoonotic fatal human respiratory CoVs [severe acute respiratory syndrome, Middle East respiratory syndrome, severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2)] and reverse zoonosis of SARS-CoV-2 to animals. A highlight of my early research was the discovery of the gut–mammary gland–sIgA axis, documenting a common mucosal immune system. The latter remains pivotal to designing maternal vaccines for passive immunity in neonates. Our discovery and innovative cell propagation of fastidious human and animal rotaviruses and caliciviruses and their infectivity in germ-free animals has provided cell-adapted and animal disease models for ongoing virologic and immunologic investigations and vaccines. Nevertheless, besides the research discoveries, my lasting legacy remains the outstanding mentees who have enriched my science and my life.

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BRIEF BIOGRAPHY

Farm Background

I grew up on a small farm in Ohio with an assortment of feathered and furry animals, besides the usual livestock, horses, and ponies, hayfields, and vegetable and flower gardens. Our farm was in the shadow of The Ohio State University (OSU), which I would later attend and where I would spend my research career as a faculty member and long-term Buckeye. I have always been interested in animals and their care and health. I was active in the local 4-H club during the summers, when I wasn't busy with routine chores (gardening and animal care) or wading in and exploring the ecology of the creek behind our house. My sister, Marilynn, and brother, Roger, and I were the first generation in my family to attend college. My parents grew up on farms in southern Ohio (Marietta), and unfortunately, neither had the means to attend college. We were a multigenerational family, with both my grandmother and great grandmother living with us, so I have many fond memories of my devoted and hard-working grandmothers and their homemade German Kaffeekuchen. Both spoke "Pfälzer" Deutsch, so I learned some German from them and more in college, which was extremely helpful when I spent a summer as an undergraduate with my relatives (none of whom spoke English) in a rural village in the Rheinland Pfälz in Germany.

Education and Career Challenges

I attended a small liberal arts college in Ohio (The College of Wooster, COW) and graduated with a BA degree in biology in 1969. Independent study (IS), the hallmark of a Wooster education, influenced my career path, even in unforeseen ways. First, I met my future husband, Mo Saif, while doing my IS in the Center for Animal Health (CFAH, formerly Food Animal Health Research Program), Veterinary Science Department at the Ohio Agricultural Research and Development Center (OARDC), now part of OSU, in Wooster. He is a DVM, PhD expert in poultry diseases, who also has had a highly successful research and administrative career in CFAH at the OARDC/OSU (now professor emeritus). I am grateful to him as my early mentor, most forthright critic, long-term partner, and ardent supporter, even after 54 years of marriage. Second, by working on my IS project in an actual research lab at the OARDC, I learned how exciting and challenging scientific research can be, especially in the field of infectious diseases. I remain a strong proponent of hands-on undergraduate research experiences to introduce talented students to future career opportunities. Many of my best graduate students first worked in my lab as undergraduates during their IS programs at the COW. As I did then, I still find lab research exciting, stimulating, rewarding, and sometimes frustrating, but always new and challenging.

After I graduated from the COW, I continued to "think small" and decided on a career in microbiology. I received a Graduate Research Fellowship from Case Western Reserve University (1969–1970), where I worked on basic immunology with Dr. Abram Stavitsky, a well-known immunologist. My project was to explore memory antibody (Ab) responses in a rabbit model, using the Jerne plaque assay to enumerate Ab-secreting cells. Although I enjoyed basic research and it became a foundation for my future immunologic research, my true passion was integrating basic with more translational studies of immunity and developing viral vaccines for livestock. Thereafter, I married in 1970 and transferred to OSU, where my husband was a faculty researcher at the CFAH/OARDC. It was there that I first encountered career challenges, including resigning my job as a research associate after our marriage due to nepotism rules that barred me from working at the OARDC. Wooster is a small town with few other science career opportunities, so this prompted me to complete the microbiology MS graduate program at OSU (in 1971) that I had begun at CWRU, and at the same time completing a high school teaching certification program. However, during my student teaching, I realized that my true passion remained scientific research.

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Figure 1

(*a*) Plastic bubble isolator units; (*b*) view of one isolator unit with a germ-free piglet inside; and (*c*) close-up of a germ-free piglet in an isolator unit in our Gnotobiotic Animal Facility, one of only a few worldwide for derivation of germ-free large animals. The availability of germ-free pigs and calves was instrumental in my research on viral pathogenesis and vaccines, elucidating the influence of probiotics and the microbiota on enteric viral infections and vaccines and animal models for human diseases.

I was next awarded a Graduate Research Fellowship in microbiology at OSU to pursue my PhD. I completed my PhD thesis research in the Veterinary Science Department at the OARDC, receiving my PhD from OSU in 1976. My MS and PhD thesis research was on passive immunity to the CoV, transmissible gastroenteritis virus (TGEV), a devastating and deadly disease in baby pigs for which there were no vaccines or treatments. This graduate research program was serendipitous for three reasons. First, I glimpsed a real virion through an electron microscope at OARDC, an instrument that I mastered and that proved key to my subsequent discovery of several new animal viruses that remained at the forefront of my research. Second, I was fortunate to have access to germ-free pigs and calves in the Gnotobiotic Animal Facility, a unique resource in the CFAH at the OARDC and one of only a few large animal germ-free facilities worldwide (**Figure 1**). A major long-term effort has been use of germ-free animals for studies of viral pathogenesis, transmission, and immunity and the microbiome and to develop and evaluate enteric viral vaccines. Third, and most importantly, Edward Bohl, an internationally recognized expert on viral diseases of swine, fortuitously served as my MS and PhD research advisor. I am extremely grateful to him for his mentoring and for guiding me toward a career in veterinary virology and immunology. He had a reputation as a meticulous, rigorous, and dedicated scientist and a very demanding advisor. I expect that my students would say that I inherited several of these characteristics, including an expectation for the pursuit of excellence by one's students and mentees.

After my PhD and the birth of our son, Justin, in 1976, I worked part-time and completed a three-year postdoc with Edward Bohl, because I was allowed to be employed at OARDC only on external grant funds. I continued my research on immunity to TGEV and another recently discovered enteric virus in swine, rotavirus. Finally, in 1979, after the relaxation of nepotism rules, I was hired as an assistant professor in the CFAH, Veterinary Science Department at OARDC/OSU. I

have remained at OSU, advancing through the faculty ranks, and was honored to be appointed as a Distinguished University Professor in 2002. My research has focused on discovering the viral causes of diseases, including zoonoses; understanding how they cause disease; and then developing preventive vaccines to alleviate suffering in both humans and animals. My laboratory has worked extensively on enteric and respiratory viruses that cause devastating gastrointestinal and respiratory disease in children and young animals.

CORONAVIRUSES

Swine Coronaviruses

The swine enteric CoVs, historically TGEV, and then later porcine epidemic diarrhea virus (PEDV) (emerged in the 1970s) and porcine deltacoronavirus (PDCoV) (identified in 2012), cause devastating gastrointestinal disease and are especially deadly in piglets. Reagents to study immune responses to TGEV were unavailable, and the correlates of immunity to TGEV were poorly understood. Vaccines were urgently needed to prevent TGEV infections and piglet deaths. This propelled my work on passive and lactogenic immunity to TGEV and vaccines. Our discoveries on mucosal and lactogenic immunity to TGEV have provided innovative vaccine strategies not only for TGEV but also for these other emerging swine enteric CoVs.

Mucosal and lactogenic immunity and discovery of the gut–mammary gland–sIgA axis. My MS and PhD research focused on coronavirus (CoV) infections, a leading killer of baby pigs and calves for which no vaccines existed. CoVs cause gastroenteritis with subsequent dehydration and death, or they can infect the lung, causing pneumonia. For the past 45+ years, my students and I have worked and published in this area, often in the face of limited funding, because CoVs were regarded as insignificant in humans before severe acute respiratory syndrome (SARS) emerged in 2003. An exciting discovery during my graduate research was the first description of an immunologic link between the intestine and mammary gland [gut–mammary gland–secretory immunoglobulin A (sIgA) axis]. We first proposed the now-established concept of a common mucosal immune system (1, 2) (**Figure 2**). Using the enteric CoV, TGEV infection of swine as a model, we established a new paradigm (plasmablasts that secrete IgA Abs in milk originate in the gut and migrate to the mammary gland during pregnancy and lactation in monogastrics such as swine and humans) (**Figure 2**). Based on this concept, we then showed that live virus or attenuated vaccines given in multiple oral doses to pregnant sows are essential to provide protective passive immunity to nursing piglets via the milk (sIgA Abs). Use of conventional intramuscular vaccines induced only serum and transient milk IgG Abs that were not protective. Thus, mucosal vaccines are needed to protect against mucosal infections and induce lactogenic immunity, which is essential to protect the newborn at birth (3–9).

When I began my MS research project and continued new aspects for my PhD, I had no idea of the challenges that I would confront, mainly, the lack of commercial immunologic reagents available for swine. This required long days and nights alone in an ancient attic lab, equipped for size exclusion and ion exchange column chromatography, to purify swine IgM, IgA, and IgG and produce antisera to them. They were essential reagents for the assays that I developed to identify the immunologic isotypes of TGEV Abs in milk and serum (1). However, this painstaking work led to our discovery that TGEV sIgA Abs were dominant in milk, after TGEV infection or attenuated oral vaccines, in contrast to IgG TGEV Abs in serum/milk after parentally administered inactivated TGEV vaccines (1–4). These breakthroughs paved the way for maternal vaccination of pregnant swine using attenuated TGEV or other enteric viral oral vaccines to induce passive lactogenic immunity via sIgA Abs in milk that protected the nursing piglets (6–9) (**Figure 2**).

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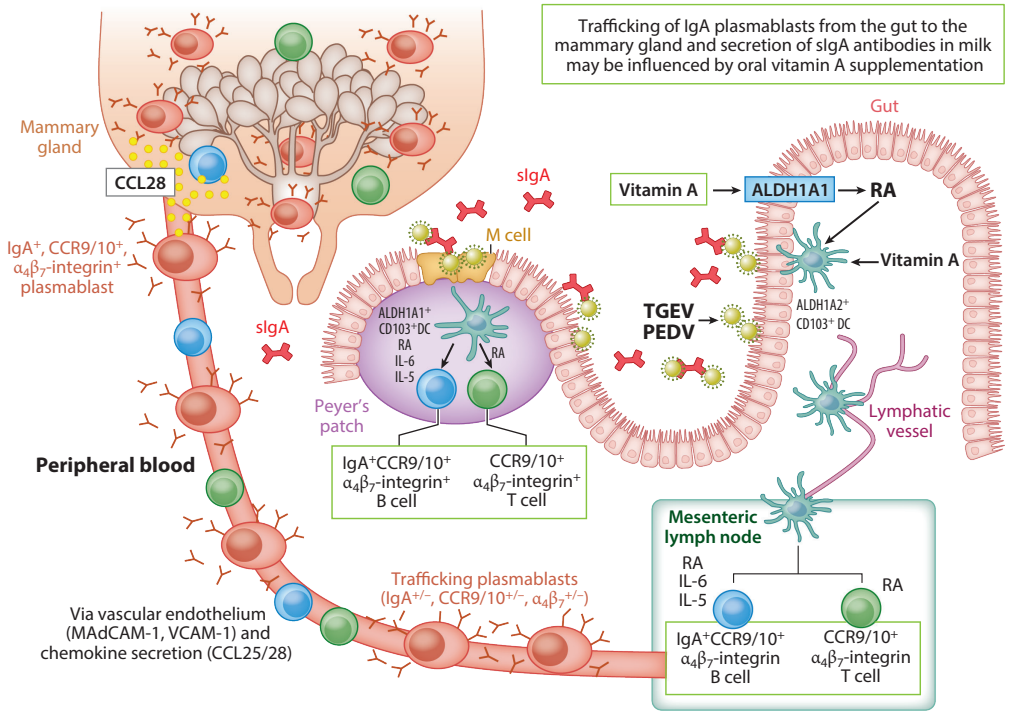

Figure 2

Gut–mammary gland–sIgA axis in sows (applicable to other monogastrics), as proposed during my MS research working on the deadly coronavirus enteric disease of swine, TGEV, and extended to other enteric coronaviruses (PEDV) and RVA (reviewed in 8, 9). Whereas infection or oral immunization of pregnant swine with TGEV induced IgA Abs in colostrum and milk that protected suckling piglets against TGEV, parental immunization with killed/subunit TGEV vaccines elicited only serum-derived IgG Abs with low, unprotective levels in milk. We proposed that viral replication in the intestine induces IgA plasmablasts in Peyer's patches and the mesenteric lymph nodes (inductive sites) and that they traffic to the gut lamina propria and other mucosal sites, including the mammary gland (effector sites), where they mature into plasma cells that secrete sIgA Abs into colostrum and milk (lactogenic immunity). These findings were the foundation for the concept of a common mucosal immune system. Later studies in mice confirmed that pregnancy hormones upregulate gut trafficking of IgA plasmablasts to the mammary gland. As depicted, gut trafficking molecules include α4β7 integrin on B/T cells that interacts with MAdCAM-1 on endothelial cells and CCR9/10 on T/B cells that interact with CCL25/28, respectively, in the gut and mammary gland. We recently extended this research to investigate the role of vitamin A on the gut–mammary gland–sIgA axis. As illustrated, gut dendritic cells possess the enzyme ALDH1A1 to metabolize dietary vitamin A to RA. The RA then upregulates the gut trafficking molecules α4β7 and CCR9/10 on primed IgA plasmablasts (26). The primed IgA plasmablasts migrate back to the gut and to the mammary gland, where they mature into plasma cells that secrete sIgA Abs in milk that provide increased lactogenic immunity to suckling neonates against enteric virus infections. The latter has been confirmed in recent studies of PEDV and RVA infections of sows given supplemental vitamin A (27–29, 30). Figure modified from original artwork by Dr. Stephanie Langel. Abbreviations: Abs, antibodies; CCL, chemokine ligand; CCR, chemokine receptor; DC, dendritic cell; Ig, immunoglobulin; MAdCAM-1, mucosal addressin cellular adhesion molecule-1; PEDV, porcine epidemic diarrhea virus; RA, retinoic acid; RVA, rotavirus A; sIgA, secretory immunoglobulin A; TGEV, transmissible gastroenteritis virus; VCAM-1, vascular cellular adhesion molecule-1.

In subsequent work, led by a highly innovative PhD student, John Van Cott, working with DVM, PhD student Theresa Brim, we used the enteric (TGEV) and the TGEV respiratory variant CoV (porcine respiratory coronavirus, PRCV) to extend the original mucosal immunity concept to show that mucosal compartmentalization exists within the common mucosal immune system, and that this affects the effectiveness of mucosal vaccines targeted to different mucosal sites (i.e., enteric versus respiratory). Thus, TGEV enteric infection protected pigs against TGEV and PRCV, but conversely, PRCV respiratory infection induced only partial immunity to TGEV (10, 11).

Additionally, two DVM graduate students, Karol Sestak and Seongkuk Park, showed that PRCV reinfection of sows in multiple pregnancies was critical to induce adequate levels of milk sIgA Abs to protect nursing pigs against TGEV (12). This cross-protective immunity likely explained how the widespread respiratory transmission of PRCV has largely displaced TGEV in swine herds, although milder TGEV variants and TGEV/PRCV recombinants continue to emerge (13). This finding is also relevant to the development of severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) mucosal vaccines that protect against both disease and virus shedding, especially if intestinal infection and fecal shedding also contribute to COVID-19 disease and virus transmission (14). In 2023, with Qiuhong Wang, a former PhD student (now a professor in CFAH and long-term collaborator), we developed a cold-adapted, temperature-sensitive, attenuated SARS-CoV-2 vaccine to elicit mucosal immunity to prevent both disease and virus shedding (15).

We have now expanded our vaccine studies aimed at protecting nursing piglets by passive lactogenic immunity to other emerging swine enteric CoVs. The devastating porcine epidemic diarrhea virus (PEDV) first emerged in the United States in 2013, killing an estimated 7 million baby pigs (16), followed in 2014 by porcine deltacoronavirus (PDCoV), another viral infection causing diarrhea in piglets (17). To control these new CoV diseases, in collaboration with a visiting scholar, Tomoichiro Oka, and with Qiuhong Wang, we first adapted PEDV to cell culture (18). Kwonil Jung, a veterinary pathologist and long-term research scientist with me, and postdocs Chun-Ming Lin and Thavamathi Annamalai characterized PEDV (19) and PDCoV (20) pathogenesis and immunity in piglets. Then, directed by Anastasia Vlasova, a former postdoc/research scientist in my lab (now an associate professor in CFAH and a long-term collaborator), we sequenced and analyzed PEDV variants and PDCoV strains circulating in swine and the related DCoV strains in birds (21, 22). Led by Qiuhong Wang and her lab, we are using reverse genetics systems to understand the genetic basis for virulence and tissue tropism (23, 24) and to develop maternal-attenuated vaccines to passively protect neonatal piglets against deadly PEDV and PDCoV infections (7–9). Finally, decades later, a similar approach (respiratory syncytial virus subunit vaccination of pregnant women) has been approved recently to provide infants with passive immunity against often fatal respiratory syncytial virus infections (25).

Vitamin A as a maternal vaccine adjuvant to enhance passive immunity in nursing piglets. Recent studies have delineated a major role for vitamin A (VA) and its metabolite retinoic acid in mucosal immunomodulation and enhanced immunity through increased sIgA Ab responses and expression of gut-homing receptors for trafficking of B and T cells to the gut (26). Based on these findings, we have explored VA deficiencies during pregnancy/lactation and a possible role for supplemental VA to optimize the gut–mammary–sIgA axis and the efficacy of mucosal and maternal vaccines (**Figure 2**). This research was conducted by teams led by two PhD students, Stephanie Langel and Juliet Chepngeno, for their thesis research and DVM, PhD postdoc Joshua Amimo and in collaboration with Anastasia Vlasova. Stephanie demonstrated that daily oral VA supplementation to PEDV-infected primiparous gilts during late pregnancy and lactation increased the numbers of PEDV IgA Ab-secreting cells and Abs and $IgA^{+}\beta7^{+}$ cells (mucosal homing) in milk, which correlated with enhanced passive lactogenic Ab protection of PEDV-challenged nursing piglets (27). We used a similar strategy in related studies, providing VA supplementation to dietary VA–deficient sows that were infected with rotavirus A (RVA) or PEDV and their piglets challenged with the respective virulent viruses shortly after birth to evaluate lactogenic immunity. Juliet found that VA supplementation of VA-deficient sows during gestation and lactation enhanced RVA-seropositive sow immune responses (anamnestic responses to RVA) (28, 29) and similarly enhanced primary immune responses to PEDV in PEDV seronegative sows (30). Passive immunity was associated with increased virus-specific IgA/IgG Abs or Ab-secreting cells in

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blood, milk, and mucosal tissues of the VA-supplemented VA-deficient sows and passive protection of their offspring (9, 27–29). We concluded that VA supplementation could provide adjuvant effects for oral-attenuated vaccines via the gut–mammary gland–sIgA axis to improve lactogenic immunity (**Figure 2**). A future goal will be to clarify the mechanisms for VA's myriad effects on mucosal immunity and vaccines.

Bovine Coronaviruses: Diversity, Diseases, and Immunity

Bovine CoVs (BCoVs) are betacoronaviruses (like SARS-CoV-2) that infect the respiratory and gastrointestinal tracts, causing pneumoenteric infections in cattle and wild ungulates (31). Intriguingly, BCoV is associated with three clinically distinct syndromes in cattle: calf diarrhea, winter dysentery (WD) in adults, and respiratory disease in cattle of various ages (31–33). We initially investigated BCoV calf diarrhea and respiratory disease using our germ-free calf model to understand viral pathogenesis and field studies to investigate their epidemiology and immunity (6, 31, 34, 35). Using field isolates from calves with diarrhea or respiratory disease to infect germ-free calves, we invariably detected diarrhea, intestinal infection with villous atrophy and fecal shedding, and consistent nasal shedding but variable respiratory disease (20–30% of calves) (36, 37). We concluded that, like other CoVs, BCoVs from calves are quasispecies, with some viruses potentially more adapted for replication in respiratory versus intestinal sites, and that variable stress factors (38) (see below) in the host may contribute to BCoV pneumonia or severe disease.

DVM, PhD student Robert Heckert conducted longitudinal studies of dairy calves from birth to 20 weeks of age. He documented both fecal and nasal shedding of BCoV, but with diarrhea being prominent in the initial infection (34, 35), followed by recurrent or intermittent nasal shedding in the same animal, with or without respiratory disease. Transient increases occurred in serum and nasal Ab titers, consistent with reinfection. These important findings, published in the veterinary literature in 1990–1991, suggest a lack of long-term mucosal immunity in the upper respiratory tract after natural respiratory BCoV infection, confirming similar observations for endemic human CoVs, and years later for SARS-CoV-2 respiratory reinfections or messenger RNA vaccines (14, 39–42).

WD in adult dairy and beef cattle and in captive wild ungulates is an acute disease characterized by severe hemorrhagic diarrhea, frequent respiratory signs, and decreased milk production in dairy cows. Its etiology was unknown for many years. Then, in 1988, we discovered BCoV in the feces of dairy cattle with WD and in the serum of the recovered cattle (31, 32, 43). Many studies followed to isolate and characterize the virus biologically (38, 44), study its pathogenesis and cross-protection against other BCoV strains (36), and investigate its epidemiology in dairy cattle (45). These studies were conducted by two DVM, PhD students (David Smith and Zacharia El-Kanawati) and two postdocs (Hiroshi Tsuemitsu and K.O. Cho) and assisted by faculty collaborators (David Benfield, D.R. Redman, and E.M. Kohler). Later, K.O. Cho also identified BCoV in feedlot cattle with WD (46). However, aspects of the disease manifestations, including severe bloody diarrhea, predilection for adult cattle, and winter prevalence, remain enigmas.

The bovine respiratory disease complex (BRDC, or shipping fever) is a costly, multifactorial disease of 6–10-month-old feedlot cattle consisting of interactions among viral, bacterial, and environmental or host stress factors that culminate in respiratory disease. Research in our lab and by others first revealed that respiratory BCoV infections play an inciting role in the BRDC, along with the other established bovine respiratory viruses (47–51). Our initial studies documented both nasal and fecal shedding of BCoV in calves shortly after arrival in feedlots, following the stress of shipping or comingling cattle. We also found an association between nasal shedding of respiratory BCoV and respiratory disease/reduced growth (48) and that a high percentage of feedlot cattle

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seroconverted to respiratory BCoV post feedlot entry. Importantly, cattle arriving with relatively high serum Ab titers to respiratory BCoV were less likely to shed BCoV, seroconvert, or develop BRDC (50, 51). This suggests that high serum Ab titers coincide with protection against BCoV respiratory infection. Nevertheless, no BCoV vaccines are licensed to prevent respiratory BCoV infections in cattle, and the correlates of immunity are unclear. Teams of DVM, PhD students (Sarah Lathrop, Armando Hoet, and Chris Thomas) and postdocs (Mustafa Hasoksuz and K.O. Cho) led and contributed to these timely and key findings.

The diversity of BCoV disease manifestations in distinct age groups prompted us to examine how these virus strains differed biologically, antigenically, and genetically. This required us to isolate these fastidious viruses in cell culture, develop antisera and monoclonals against the virus isolates from the different clinical entities, and conduct in vitro cross-neutralization in cell culture and in vivo cross-protection studies in germ-free calves. No consistent antigenic markers were identified to discriminate BCoVs from the different clinical syndromes. All BCoVs belonged to a single serotype based on virus cross-neutralization tests and calf cross-protection studies, although 2–3 subtypes of BCoV were recognized based on monoclonal Ab analysis (36, 52). However, the calf cross-protection experiments revealed complete cross-protection against homologous strains but subclinical virus shedding (by real-time polymerase chain reaction) in calves challenged with the heterologous BCoV strains (36). These data confirmed field studies documenting subclinical BCoV infections (34, 35) and the potential for subclinically infected animals to be BCoV reservoirs in infected herds. These key discoveries resemble findings for SARS-CoV-2 in humans, documenting asymptomatic infections and partial immunity to variants (40, 41).

Sequence analysis of the full genomes of pairs of respiratory/enteric BCoV isolates from feedlot cattle by postdoc Xinsheng Zhang and collaborators from OSU and The Institute for Genomic Research yielded several important findings. There were no consistent genetic markers to discriminate between respiratory and enteric BCoVs; genetic diversity between contemporary and historic BCoV strains was greater than between contemporary enteric and respiratory strains; and curiously, in the process of cell culture adaptation, an enteric BCoV strain accumulated mutations to resemble the corresponding respiratory BCoV isolate from the same animal (53). Infectious clones of BCoV (a future aim) are critical to elucidate whether there is a genetic basis for the varied BCoV disease syndromes and tissue tropisms in concert with host factors.

These are critical findings to guide vaccine development and suggest that a single broadly cross-reactive strain of BCoV may suffice for a broad-spectrum vaccine against all three clinical syndromes. Our accumulated data on BCoV infections have many corollaries to SARS, Middle East respiratory syndrome (MERS), and SARS-CoV-2 infections of humans, as discussed in detail in other reviews (6, 14, 31). Many dedicated graduate students and postdocs worked on the multiple and long-term aspects of this BCoV research, as identified in the prior sections, but also including graduate students K.O. Chang, Paul Nielson, and Musaad Al-Dubaib.

WHO DISEASE X AND THE COMING PLAGUES

Interspecies Transmission of Bovine and Other CoVs: Precedents for the Zoonotic Emergence of Fatal Human CoVs (SARS, MERS, SARS-CoV-2)

For much of my career, there was little funding for or interest in animal CoVs outside the veterinary community. Then, from Guangdong Province in China in 2002–2003, a new disease, SARS, emerged from bat reservoirs, spilling over to civet cats in the wet markets and then as a zoonosis to humans in China [see the review by our Independent Task Force on SARS-CoV-2 (54)]. A decade later, in 2012, the ongoing MERS CoV spilled over from camels to humans in the Middle East. Since 2019, only seven years after MERS was first reported in humans, we are still recovering

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from the precipitous global spread of SARS-CoV-2 that ignited a devastating and ongoing pandemic (54). SARS first showed us that diseases may be invisible passengers on airplanes, so that a new disease emerging in China may cross oceans and continents to resurface in Canada or the United States within the time it takes a plane to circumnavigate the globe. Suddenly, our years of painstaking research on animal CoVs provided the foundation for an understanding of these newly emerged viral pathogens in humans (6, 31, 55–57). Global efforts, integrated One Health strategies, and a new generation of dedicated transdisciplinary scientists are needed to combat these emerging zoonotic diseases.

Prior to the emergence of SARS-CoV-2, there were many unheeded precedents for interspecies transmission of CoVs. In 1995, postdoc Hiroshi Tsunemitsu, and later another postdoc, MyungGuk Han, confirmed the interspecies transmission of CoVs among cattle; cervids, including white-tailed deer (58); and humans (59) using our unique germ-free calf model. In collaboration with my husband Mo and his student Mahmoud Ismail, we confirmed BCoV transmission even to avian species (60). Thus, contrary to the prevailing dogma at the time, CoVs are not species specific; they infect farm animals and wildlife and can transmit among these species, sometimes spilling over into humans. We isolated each of the viruses in cell culture and characterized their antigenic relatedness. After finally procuring funds to sequence multiple ~30-Kd CoVs (which was extremely expensive then), two postdocs in my lab, Konstantin Alekseev and Anastasia Vlasova, in collaboration with David Spiro (the J. Craig Venter Institute) and Rebecca Halpin and coworkers (University of Pittsburgh), analyzed the full genomes of the closely related CoVs from the wild ruminants (cervids) (61) in comparison with those from cattle (WD, calf diarrhea, and BRDC strains) (53). These viruses and the human strain from a child with diarrhea (62) had >98% sequence identities in the full genomes (S/HE for the human strain), and antisera cross-neutralized the viruses. This indicated that, along with BCoV, they are a single species in the *Embecovirus* subgenera and likely represent host-range mutants of a common ancestral CoV that cocirculate and are maintained in an ungulate reservoir community.

A similar broad host range is exemplified by SARS, MERS, and COVID-19, which are likely zoonoses caused by newly emergent CoVs (54). To survey animals as potential reservoirs or intermediate hosts for zoonotic CoVs, new and specific serologic assays were needed to detect Abs in the various animal species. Cross-reactivity among CoVs hampered use of serologic tests such as ELISA (enzyme-linked immunosorbent assay). We observed unexpected cross-reactivity between the SARS betaCoV and animal alphaCoVs (TGEV). Postdoc Anastasia Vlasova, in collaboration with Lia Haynes from the US Centers for Disease Control and Prevention (CDC), generated SARS-CoV-specific recombinant N protein and developed a SARS-CoV-specific Ab ELISA assay for screening human and animal sera to detect animal hosts for SARS (63). In collaboration with international scientists (Malik Peiris, Leo Poon, and colleagues at Hong Kong University and M. Hemida et al. in the Kingdom of Saudi Arabia) and with OSU collaborators (led by the Scott Kenney and Shan-Lu Liu labs and Anastasia Vlasova with PhD student Bindu Adhikari), we developed and validated similar MERS- and SARS-CoV-2-specific Ab tests for serologic testing of suspect animal reservoirs, including deer; testing of breakthrough variants in humans that escape immunity; and evaluation of cross-reactivity with endemic human CoVs (42, 64–68).

Alarmingly, SARS-CoV-2 is a generalist virus that infects (without initial notable genetic changes) not only humans but at least 34 animal species (54). Two OSU collaborators (Vanessa Hale and Andrew Bowman), with our wildlife team (eScout), the OSU Infectious Diseases Institute (Seth Faith and Mike Oglesbee), and our lab, led an investigation of SARS-CoV-2 in Ohio wildlife. We were first to detect SARS-CoV-2 in 36% of screened white-tailed deer and demonstrate their repeated spillover from humans to deer and deer-to-deer transmission, validating deer as a potential wildlife reservoir for SARS-CoV-2 (69). This has now been documented in multiple

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studies of US and Canadian deer, raising concerns of whether the virus will be maintained in this wildlife host with spillback to humans or other animals (54).

Besides ungulates, pet animals may also be hosts for interspecies and zoonotic transmission of CoVs. There is evidence for SARS-CoV-2 transmission from humans to pets, and vice versa (reviewed in 54). Likewise, Anastasia Vlasova and our labs, in collaboration with Greg Gray at University Texas Medical Branch and colleagues from Malaysia, characterized a new strain of recombinant canine alphaCoV (CCoV), designated HuPn-2018, from children with pneumonia in Malaysia (70). How widespread this recombinant CoV is in humans or dogs is unknown, but a closely related strain (>99% genome identity) was detected in adults returning from Haiti (71). This and our earlier research (in collaboration with a former PhD student, Marli Azevedo, at the US Food and Drug Administration) identifying feline-like alphaCoVs in patients with pneumonia in the United States raise concerns for pets as vectors, with at least occasional alphaCoV-1 spillover to humans (72).

In 2015, with a postdoc from China (Hui Hu), Kwonil Jung, and others at CFAH, we were first to isolate the emerging PDCoV from pigs in the United States (73). With Scott Kenney, Kwonil Jung, and collaborators from the Netherlands led by Berend Jan Bosch, we showed that PDCoV efficiently infects human cells via an interspecies conserved site on aminopeptidase N receptor (74) and experimentally infects poultry (75) and calves (76). This broad host range may facilitate PDCoV transmission to non-reservoir species, including children, as documented recently in Haiti (77). Thus, both human CCoV and human PDCoV should be monitored closely for evidence of efficient human-to-human transmission and pandemic potential, suggestive of emerging World Health Organization (WHO) Disease X (54).

Bovine and Porcine CoV Infections in Their Native Hosts as Models to Understand the Pathogenesis and Transmission of the Emerging Fatal Human CoVs (SARS, MERS, SARS-CoV-2)

When SARS-CoV first emerged in humans in 2003, there were few relevant animal models to understand the pathogenesis of respiratory CoVs. Two endemic respiratory CoV infections of livestock, BCoV and PRCV, share several features with SARS-CoV and SARS-CoV-2. BCoV has a broad host range, including wild ungulates, and zoonotic potential (see prior section). It also has a dual tropism for the respiratory and gastrointestinal tracts. These aspects, their interspecies transmission among multiple species, and factors that impact disease severity in cattle parallel related facets of SARS-CoV or SARS-CoV-2 in humans (reviewed in 31).

Because of its infection of the lung and the atypical pneumonia it induces, resembling that in SARS and SARS-CoV-2 patients, PRCV infection of pigs provided an excellent model to discern the factors that enhance pathogenicity. Kwonil Jung and postdoc Xinsheng Zhang, with input from postdocs Konstantin Alexseev and Anastasia Vlasova and OSU colleague Aradhya Gourapura, investigated these factors in a PRCV pig model (reviewed in 31). Dual infection with another swine respiratory virus (porcine reproductive and respiratory syndrome) preceding PRCV infection reduced weight gains and increased pneumonia severity compared with either single infection (78). Because SARS and SARS-CoV-2 patients may be coinfected with other human respiratory pathogens, such animal models are crucial to ascertain their dual impacts. Additionally, many SARS and COVID-19 patients were treated with corticosteroids, but the potential secondary effects are unclear. Treatment of PRCV-infected pigs with corticosteroids alleviated PRCV pneumonia early, but continued use exacerbated later stages of infection, possibly via the decreased cellular immune responses observed in the lungs, creating an environment for more extensive viral replication (79). Likewise, recrudescence of BCoV fecal shedding was observed in one of four WD BCoV-infected cows treated with dexamethasone (38).

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ROTAVIRUSES

Dual Research to Understand Rotavirus Pathogenesis and Develop Animal and Human Rotavirus Vaccines: The Germ-Free Pig Model

The enteric virus rotavirus genogroup A (RVA) is recognized as the leading cause of gastroenteritis in infants and young animals worldwide. An estimated 200,000 children die of rotavirus diarrhea annually, but human RVA oral vaccines are not highly effective in impoverished lower-middle-income countries (LMICs) where they are most needed. Similarly, commercial RVA vaccines for young pigs and calves lack efficacy (80). The lack of cell culture–adapted strains for in vitro assays and vaccine development hampered rotavirus research for many years. In conjunction with colleagues at OSU and Richard Wyatt, Al Kapikian, and others at the National Institutes of Health (NIH), we were the first to grow human RVA in cell culture after first passaging the virus (in a human stool sample) in a germ-free piglet to remove interfering Abs and amplify virus titers (81). Multiple human RVA strains infected and caused diarrhea in the gnotobiotic pig model, and it remains the only animal model with prolonged susceptibility to human RVA disease (reviewed in 82–86).

Subsequently, using this unique animal model, and with Taka Hoshino and NIH coworkers, we used reassortant human/porcine RVA strains to identify the RVA genes involved in infection, virulence, and attenuation (87). PhD student Yusheng Guo, working with Anastasia Vlasova and my lab, sequenced pairs of virulent and attenuated human and swine RVA stains from our lab and identified two mutations in RVA outer-capsid VP4 that may affect RVA attenuation (88). Yusheng further confirmed that differential histo-blood group antigen (HBGA)–RVA interactions through VP4 determine replication efficacy of virulent RVAs in porcine intestinal enteroids (89). Now with John Patton at the University of Indiana, we are using a RVA reverse genetics system to define the genes related to virus attenuation and replication. These breakthroughs established the relevance of the germ-free piglet model for human RVA that we have used extensively to design and test infant and piglet RVA vaccines, to understand neonatal mucosal immunity and immune correlates of protection, and to define the genetic basis for RVA attenuation/virulence (8, 82–86). Our work facilitated the development of the current attenuated RVA vaccines used orally in infants to elicit active immunity. We have subsequently developed multiple approaches to enhance the effectiveness of such vaccines, as summarized below, with the ultimate goal of improving RVA vaccines to protect infants in LMICs as well as nursing and weaned pigs and calves on the farm.

The objectives of our early work using the germ-free pig model of human RVA infection were to understand the correlates of protection to an attenuated (by serial passage in monkey kidney cells) oral human RVA vaccine (AttRVA) (81). The corresponding virulent RVA strain maintained by germ-free pig passage was used for challenge. These initial studies revealed that intestinal IgA RVA Abs and Ab-secreting cells, and RVA-specific memory B cells and IFN-γ-secreting T cells induced after virulent RVA infection, correlated with protection against RVA challenge (8, 82–84, 90). At least two to three oral doses of AttRVA induced moderate protection. Prime-boost strategies of nonreplicating RVA inner-capsid viral-like particles (VLPs) 2/6 VLP oral or intranasal (IN) vaccines with *Escherichia coli* heat-labile enterotoxin (mLT) adjuvant followed by oral AttRVA booster were ineffective for priming mucosal immunity and inducing protection. The converse, oral AttRVA priming and IN 2/6 VLP+mLT boosting, was effective, confirming that replicating live virus was needed to prime intestinal IgA and RVA-specific IFN-γ-secreting T cell responses and mucosal immunity (reviewed in 8, 85, 86). Using the latter regimen and substituting immunostimulating complexes as a mucosal adjuvant was also effective (85). In further studies of mucosal IgA memory B cell responses to human RVA in piglets, we discovered another original immunologic finding: Effector IgA memory B cells reside in the gut and spleen and not in the

bone marrow of pigs (91). The latter concept explained our findings that in TGEV (92) or RVA naturally infected or orally vaccinated pigs (5), parenteral or IN subunit booster inoculations could enhance mucosal IgA Ab responses. This paved the way for new prime/boost mucosal vaccination strategies, now used in the field to control endemic virus infections. These pioneering studies were conducted by teams of dedicated and talented PhD students and postdocs, including Marli Azevedo, Ana Gonzalez, Annelise Geyer, Christina Iosef, Trang Van Nguyen, Blair Rosen, Lucy Ward, Lijuan Yuan, and Zhenwen Zhang.

Because the efficacy of RVA oral vaccines is reduced in women or swine with RVA maternal Abs, we investigated the mechanisms in our germ-free piglet model. We noted that maternal RVA Abs suppress effector and memory B cell responses to oral RVA and to the above prime/boost vaccine regimen, thereby decreasing active RVA Ab responses (8, 93–95). Given RVA vaccine failures in infants in LMICs, we tested alternative passive Ab treatments in the germ-free piglet model to reduce RVA diarrhea. Both egg yolk IgY RVA Abs and llama variable heavy chain (VHH) Ab to RVA Vp 6 provided passive Ab protection that prevented RVA diarrhea (96, 97). It is still unknown how non-neutralizing Abs to RVA inner-capsid protein VP6 efficiently boost RVA immunity (2/6 VLP) after AttRVA oral priming or provide passive immunity (VHH Vp6 Ab). Further research to decipher the mechanisms is needed. These studies were led by research scientist Doug Hodkins and teams of PhD students and visiting scholars, including Marina Bok, Kuldeep Chattha, Louisa deArriba, SY Kang, Vivianna Parreno, Thanh To-Long, Celina Vega, Lucy Ward, Lijuan Yuan, and other coauthors.

Novel Immune Modulators (Vitamin A, Probiotics) for Improved Oral RVA Vaccines to Elicit Active Immunity in Infants in LMICs and Neonatal Animals

Because of failures of RVA oral vaccines in LMICs and in neonatal animals, we are developing innovative, cost-effective strategies (VA, probiotics) to enhance host immune responses and vaccine efficacy and evaluating the selected scenarios in the germ-free piglet model. We discovered that prenatal VA deficiency impairs neonatal innate and adaptive immunity to RVA in multiple ways and compromises the effectiveness of RVA oral vaccines, including that of a licensed commercial human RVA vaccine (8, 86, 98–100). Unfortunately, supplemental VA administered to neonates with three-dose oral AttRVA vaccine failed to compensate for the prenatal VA deficiency or moderate the dysregulated immune responses, suggesting that longer-term or maternal VA supplementation in pregnant or lactating women may be required.

Our probiotic studies have revealed that colonization of germ-free pigs with selected lactic acid bacteria (LAB) enhances immune responses to AttRVA oral vaccines in neonates (8, 101, 102). Moreover, colonization by combinations of probiotics (LAB with bifidobacteria common in breastfed infants) had a greater impact than LAB alone in moderating RVA diarrhea or enhancing immunity to RVA vaccines. More recently, we discovered that the probiotic *E. coli* Nissle 1917 more effectively enhanced immune responses to AttRVA vaccines and moderated RVA diarrhea in neonates via multiple mechanisms, including by direct binding to RVA particles (101, 103, 104). Our unique germ-free piglet model and innovative studies of the selected probiotics have improved understanding of maturation of neonatal mucosal immunity and provided new insights into the beneficial effects of probiotics/commensals to alleviate diarrheal disease and enhance oral vaccines (8, 101, 102).

Malnutrition and gut microbiota dysbiosis are common in infants in LMICs, where oral RVA vaccines have reduced efficacy. Recently, we developed a protein-deficient germ-free piglet model stably colonized with infant fecal microbiota to investigate the impact of malnutrition and the infant microbiome on RVA immunity, vaccines, and disease. We demonstrated that neonatal protein

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deficiency results in immune suppression that increases RVA disease severity and decreases RVA vaccine efficacy (105, 106). Our prior and ongoing studies will unravel interactions among the human gut microbiota, viral pathogens or vaccines, diet, and immunomaturation in neonates and investigate nutritional and probiotic interventions (107–110). Further, our findings will suggest new strategies to enhance oral viral vaccines and limit viral diarrheas in infants in LMICs through short-term coadministration of probiotics/commensals to mothers or breastfed infants. These timely and innovative studies were co-led by myself and Anastasia Vlasova and conducted by teams of dedicated and talented postdocs and students, including Sukumar Kandasamy, Kuldeep Chattha, David Fischer, Ayako Miyazaki, Michael Husheem, Stephanie Langel, Francine Chimelopaim, Juliet Chepngeno, Lulu Shao, Moyasar Alhamo, Joshua Amimo, and Sergei Raev, as well as faculty collaborator Gireesh Rajashekara and his lab members.

Bovine Rotavirus Maternal Vaccines to Passively Protect Nursing Calves

A highlight of my career and research on RVA immunity and vaccines was a short-term Fulbright Fellowship at the Institute of Virology at CICV-INTA in Buenos Aires, Argentina, in 1995. I did not anticipate that my visit would initiate a long-term research collaboration with my Argentine colleagues, as well as the recruitment of a cadre of outstanding students, postdocs, and faculty collaborators. The latter included the former directors of the Institute of Virology, Alejandro Schudel and later Ana Sadir and Fernando Fernandez, and students Alejandro Lucchelli, Vivianna Parreno, Celina Vega, and Marina Bok, who worked on RVA projects. The initial research included bovine RVA vaccine field trials in herds with >10,000 cattle in the Argentine Pampas. I was fortunate to visit the cattle estancias and experience the impact of RVA vaccines firsthand on horseback with the gauchos (**Figure 3**).

Earlier we had established that most adult cattle are RVA seropositive but that Abs in colostrum/milk decline to unprotective levels for the nursing calves (5, 80). Unlike in monogastrics, where IgA Abs dominate in milk (**Figure 2**), in ruminants, IgG1 Abs are dominant because they are selectively transported via the FcRn from serum into colostrum and milk (80). We first isolated and characterized RVA strains from the animals, identifying new porcine and bovine RVA serotypes/genotypes (111, 112), and designed the appropriate maternal vaccines and vaccination

Figure 3

My research on bovine rotavirus A (RVA) vaccines led to a Fulbright Fellowship in Argentina to conduct RVA vaccine trials in large beef herds. On horseback, I experienced directly the impact of RVA maternal vaccines in cows on passive protection of their calves against RVA diarrhea. This photo shows me with the gauchos on the Estancia of Ingeniero Romat in the Argentine Pampas.

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routes (based on the required Ab isotypes in milk) to passively protect nursing piglets (prior sections) and calves (5, 80). We discovered that use of baculovirus-expressed RVA triple capsid 2/4/6/7 VLPs (containing RVA outer-capsid proteins VP4 and VP7 that induce neutralizing Abs) with oil adjuvants were effective as parenteral maternal booster vaccines in RVA-seropositive cattle to increase IgG1 Abs to protective levels in colostrum/milk (113, 114). Doug Hodkins and Fernando Fernandez led this research. Besides the Argentine students and postdocs mentioned, many OSU students in my lab contributed to these studies (Yunjeong Kim, K.O. Chang, Paul Nielson, and Anil Parwani), along with long-term collaborators from Baylor University in Houston, Texas (M. Conner, S. Crawford, and M. Estes). Thus, our work has facilitated the development of the current attenuated RVA vaccines used in pregnant farm animals to passively protect nursing neonates in Argentina, the United States, and elsewhere.

Discovery of Novel RV Genogroups from Swine and Cattle Closely Related to Human RVs

Besides our focus on RVA immunity and vaccines, in 1980 we discovered the first non-RVA detected from any species, porcine rotavirus C (RVC), the first member of a new RVC genogroup (115, 116). It was later confirmed to be closely related to similar RVC strains causing diarrhea in children (117) and calves (118). In further studies of RVB and RVC strains (reviewed in 119 and 120), we confirmed their intestinal pathogenesis in swine and cattle (121, 122), sequenced and analyzed their genetic diversity (122–124), used novel approaches to adapt RVC to cell culture (118, 125), and produced VLPs for studies of immunity and vaccines (126–128). However, the RVBs remain refractory to propagation in cell culture, hindering further in vitro investigations. Besides the mentees mentioned earlier, students Baoming Jiang (now at the CDC) and Lisa Terrett and Ken Theil (retired faculty in CFAH) contributed greatly to these findings.

CALICIVIRUSES INCLUDING HUMAN NOROVIRUSES

There are no vaccines or specific treatments for most caliciviruses, including noroviruses (NoVs), the leading cause of foodborne disease worldwide. This virus was first discovered in Norwalk, Ohio, in 1972 by Al Kapikian at NIH, and it remains a leading cause of gastroenteritis (129). We discovered that caliciviruses also cause diarrhea in swine and cattle, and like for animal rotaviruses, germ-free pigs and calves can be used as models for the corresponding human disease (120, 130, 131). Our research has focused on discovering and characterizing such novel enteric viruses and their adaptation to growth in cell culture, a first step in attenuated vaccine development or rescue of infectious clones. In 2002, postdoc Jeff Smiley was first to sequence the NB strain of bovine calicivirus from diarrheic calves in the United States, which was designated as a member of a new genus of calicivirus (131). Later, after sequencing of the bovine Newberry virus in the United Kingdom, both strains were designated as members of the new *Nebovirus* genus (132). We further characterized bovine NoVs from cattle, their pathogenesis (133), their cross-reactivity with human NoVs and neboviruses (134, 135), and their prevalence in cattle (136).

Our lab discovered the first porcine enteric calicivirus, a new sapovirus that was related to similar viruses causing diarrhea in children (115), and later identified additional new caliciviruses in swine (120, 130). In further studies, with students/postdocs Bill Flynn, Mingzhang Guo, Anil Parwani, MyungGuk Han, Qiuhong Wang, Viviana Costantini, Zhongyan Lu, Tomoichiro Oka, Zufan Sisay, Sayaka Takanashi, and others, we confirmed the intestinal pathogenesis of the virus (137) and its genetic variability and prevalence in swine (138). We used novel approaches to adapt it to cell culture, confirmed the molecular basis for virulence and cell adaptation (139, 140), and produced attenuated viruses and VLPs for candidate vaccines (139–141). We established a novel

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system to cultivate fastidious enteric caliciviruses by adding intestinal contents from germ-free pigs to the medium (142, 143). Subsequently, in collaborative studies with Kim Green at NIH and her postdoc K.O. Chang (my former PhD student), we identified bile acids as the component of intestinal contents that enhances enteric calicivirus replication (144). This discovery revealed new avenues for propagation of other non-cultivable caliciviruses in cell cultures or enteroids (human sapoviruses and NoVs) (145, 146) and a novel mechanism (via bile acids) exploited by enteric viruses to replicate in the host upper small intestinal tract.

Swine caliciviruses continue to serve as models to study the related human sapoviruses and NoVs, the leading causes of foodborne infections. Students/postdocs instrumental in the following studies included Sonia Cheetham, Minera De Souza, Thavamathi Annamalai, and Kwonil Jung. To develop a human NoV animal model, we discovered that swine express HBGAs similar to those of humans on gut tissues, and as in humans, the HBGA types were related to susceptibility to infection (147). Intriguingly, intestinal colonization of germ-free pigs with a commensal, *Escherichia fergusonii* (which expresses HBGA A and H), promoted infectivity of human NoV GII.12, possibly by enhancing attachment of human NoV to enterocytes (148). More surprising, and a new paradigm for food safety, was the revelation that human NoV-VLPs also bind to related HBGA carbohydrates on plants (e.g., lettuce) and are internalized, preventing their removal by washing (149, 150). Thus, we are investigating further the stability and foodborne transmission of NoVs on fruits and vegetables and examining lectins as competitors for NoV binding to leafy greens. These latter investigations were led by Qiuhong Wang and conducted by postdoc Malak Esseili and other lab members.

OTHER SCIENTIFIC ACTIVITIES

Being a scientist has also enabled me to generate international scientific collaborations, many resulting in lasting friendships and recruitment of outstanding international students. These have included a Fulbright Fellowship to Argentina to test RVA vaccines in cattle herds (prior section; **Figure 3**). Perhaps the most challenging scientific task was my service as a scientific advisor for a US Agency for International Development project from 1990 to 1995 promoting scientific collaboration among Egyptian and Israeli scientists in controlling ruminant diseases common to both countries. A major challenge was building trust and fostering research cooperation among the previously estranged Egyptian and Israeli scientists and their associates to address and solve regional and transborder disease problems. In-person scientific meetings, first in the United States and then in Egypt and Israel, encouraged personal contacts to exchange scientific information directly to discuss and solve common ruminant disease problems. These scientific exchanges promoted ongoing and future Egyptian–Israeli research collaborations as a vital part of the project.

During the SARS, MERS, and SARS-CoV-2 outbreaks, I unexpectedly found myself at the epicenter of interest in CoVs. In 2003, I was a member of an advisory team on SARS-CoV for the CDC and Advisor to the WHO Scientific Advisory Committee on SARS-CoV. In 2014, I was a consultant on MERS CoV in camels for the Saudi Arabian Ministry of Agriculture, Ryadh, Kingdom of Saudi Arabia. Now with myself and Qiuhong Wang as co-directors, and through the OSU Infectious Diseases Institute (Mike Oglesbee, Director), we are an International Food and Agriculture Organization (FAO) Reference Center for Zoonotic Coronaviruses and part of the World Health Organization (WHO) Coronavirus Network of Reference Labs (CoViNet). Importantly, encountering the disease problems directly in the field—calves dying of RVA and CoV infections in large herds in Argentina or in Egypt, Israel, or the United States; camels with MERS CoV in Saudi Arabia; and the diverse wildlife species in overcrowded wet markets in China and Southeast Asia—enabled me to appreciate the disease complexity, interwoven with environmental and societal factors, and a need for multifaceted solutions (i.e., the One Health paradigm; see 54).

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In 2003, I was both amazed and humbled to be elected to the US National Academy of Sciences, Engineering and Medicine (NASEM). Being in the company of Nobel prize–winning scientists at the NASEM annual meeting is truly inspiring. I have been active in the NASEM. I served as a former chair for our section, and most recently, I served as a task force member on Countering Zoonotic Spillover of High Consequence Pathogens in the Southeast Asian region that has now culminated in the NASEM and INGSA (International Network for Government Science Advice) Guidebook now available online (**https://ingsa.org/**).

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